

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Supplementary Examination – 2023

Course: B. Tech. Branch : Electronics & Telecom Engg. Semester : V

Subject Code & Name: BTETOE505B/BTEXOE505B

Artificial Intelligence and Machine learning

Max Marks: 60

Date: 19-08-2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

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| A) What are the foundations of Artificial Intelligence? How the Environments in AI can be categorized. | (L1/CO1) | 6 |
| B) Explain the cognitive modeling approach and rational agent approach in AI. | (L1/CO2) | 6 |
| C) Explain the different structures of Agents and concept of Turing test. | (L1/CO1) | 6 |

Q.2 Solve Any Two of the following.

12

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| A) How the problems can be formulated in AI. Explain with an example. | (L2/CO2) | 6 |
| B) Explain Iterative deepening depth-first search and Bidirectional search in AI with advantages and disadvantages. | (L1/CO3) | 6 |
| C) What are the Heuristic Functions. Explain the effect of heuristic accuracy on performance in AI model. | (L1/CO1) | 6 |

Q. 3 Solve Any Two of the following.

12

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| A) Write and explain hill-climbing and steepest hill-climbing search algorithms in artificial intelligence. | (L2/CO2) | 6 |
| B) Write and explain the A* algorithm with advantages and disadvantages. | (L2/CO3) | 6 |
| C) Explain Alpha Beta Pruning and Move ordering concepts. | (L1/CO1) | 6 |

Q.4 Solve Any Two of the following.

12

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| A) Explain the concepts of Forward Chaining and Backward Chaining | (L1/CO2) | 6 |
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| B) | Explain Inductive Logic Programming and Statistical Learning | (L3/CO3) | 6 |
| C) | Explain the terms in model selection | (L1/CO1) | 6 |
| | 1. Complexity versus goodness of fit | | |
| | 2. From error rates to loss | | |
| | 3. Regularization | | |

Q. 5	Solve Any Two of the following.		12
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|-----------|--|----------|----------|
| A) | 1. What is the difference between supervised and unsupervised learning | (L2/CO2) | 6 |
| | 2. Write a note on linear regression. | | |
| B) | What is Ensemble Learning? What are the types of ensemble learning? | (L2/CO3) | 6 |
| C) | Write a note on Expectation-Maximization (EM) algorithm. | (L2/CO2) | 6 |

***** End *****